

Tennis Analytics Dashboard — Project Documentation

1. Introduction:

This project was developed to build a complete analytics system using real-world tennis data from the SportRadar API. Instead of treating the API as just a data source to display results, the goal here is to develop a proper end-to-end analytics workflow.

The project covers the full lifecycle of data:

- Extracting raw JSON data from an API
- Cleaning and mapping fields properly
- Designing a normalized relational schema
- Loading structured data into MySQL
- Writing analytical SQL queries
- Building an interactive Streamlit dashboard
- Generating meaningful insights from ranking and competition data

The core idea was to convert unstructured sports API data into structured analytical intelligence.

2. Project Workflow:

This project was implemented in multiple controlled stages instead of building everything at once. Each stage was tested separately before proceeding to the next.

- **Stage 1 - API Understanding & Exploration**

Before creating any database or dashboard logic, the API response structure was studied carefully.

Steps followed:

- Manually called SportRadar endpoints
- Printed raw JSON responses
- Inspected nested structures
- Identified:
 - Stable fields
 - Optional fields
 - Nested objects
 - Missing value patterns

This step was crucial because incorrect assumptions about the JSON structure caused null insertions, which were fixed only after careful inspection of the responses.

- **Stage 2 - Data Extraction Scripts (Python)**

Separate Python scripts were created for each dataset type:

- competitions_api.py
- complexes_api.py
- rankings_api.py

Design approach:

Instead of storing JSON directly into the database:

- Each record was parsed carefully
- The .get() method was used for optional fields
- Nested fields were flattened
- Foreign key relationships were maintained

- **Stage 3 - Data Cleaning & Mapping**

During extraction:

- Missing fields were handled safely
- Differences in country or city nesting were addressed
- Reserved SQL keywords were renamed
- Datatype consistency was ensured

Solution:

- Rank was renamed to ranking
- Nested venue.city.name was corrected to flat fields when the API changed structure

- **Stage 4 - Database Design & Creation**

Instead of using a single large table, the schema was normalized into multiple tables.

Reason:

- Avoid duplication
- Improve join performance
- Enable analytical queries

- Maintain referential integrity

First, we created the tables, then inserted data, and finally validated foreign keys.

- **Stage 6 – Streamlit Dashboard Development**

The dashboard was built as a multi-page app:

- Homepage
- Search & Filter
- Competitors Details
- Country Analysis
- Leaderboards
- SQL Query

Focus was placed on:

- User interaction
- KPI storytelling
- Visual clarity
- Analytical relevance

3. Schema Design:

Schema design was one of the most important parts of this project because sports data is highly relational. Instead of storing everything in one table, we explicitly modelled the relationships.

Competitors Table:

Purpose: Stores unique player identity information.

Fields:

- competitor_id - Primary Key
- name
- country
- country_code
- abbreviation

Competitor_Rankings Table:

Purpose:

Stores' ranking metrics that change over time.

Fields:

- rank_id – Auto Increment Primary Key
- ranking
- movement
- points
- competitions_played
- competitor_id – Foreign Key (Competitors Table)

Note:

Here, rank_id is set to Auto Increment instead of using the API value to avoid collision risk.

Competitions Table:

Purpose: Stores tournament structures.

Fields:

- competition_id - Primary Key
- competition_name
- parent_id - Foreign Key to same table
- type
- gender
- category_id - Foreign Key (Categories Table)

Categories Table:

Purpose: Competition classification lookup table.

Fields:

- category_id
- category_name

Complexes Tables:**Fields:**

- complex_id
- complex_name

Venues Tables:

Purpose: Store venue details.

Fields:

- venue_id – Primary Key
- venue_name
- city_name
- country_name
- country_code
- timezone
- complex_id - Foreign Key (Complexes Table)

4. Relationship Design Logic:

Relationships implemented:

Competitors → Rankings (1-to-many)

Categories → Competitions (1-to-many)

Competitions → Competitions (self hierarchy)

Complexes → Venues (1-to-many)

5. Challenges Faced:

Challenge 1 - API Structure Inconsistency

Problem:

Some fields appeared nested in documentation but flat in the actual response.

Impact:

City and country fields were inserting NULL values.

Solution:

Printed raw API JSON, remapped fields, and corrected the extraction logic.

Challenge 2 - SQL Reserved Keywords

Problem:

Using the column name rank caused syntax errors.

Fix:
Renamed to ranking.

Challenge 3 - Duplicate Insertions

Problem:

API returns repeated entities across endpoints.

Fix:

- Enforced primary keys
- Used INSERT IGNORE

6. Streamlit Dashboard:

Below are the key insights derived from each dashboard module.

a) Homepage - Dataset Scale & Structure Insights

The homepage KPIs immediately establish the scale and diversity of the dataset.

Observed metrics:

- Approximately 500 competitors
- 64 countries represented
- Highest points exceeding 8,000

Insight:

The dataset is globally distributed and not focused on a few regions. With competitors from 60+ countries, the ranking ecosystem is internationally diverse. However, further country analysis shows that the contribution is not evenly distributed; some countries perform better despite similar participation counts.

b) Competitor Details Page - Individual Performance Behaviour

The competitor drill-down page displays individual metrics for each player:

- Rank
- Points
- Movement
- Competitions played

Insights:

Rank itself is static, but movement indicates direction. A player ranked lower but with positive movement may be improving at a faster rate than a higher-ranked player with negative movement. This suggests that ranking movement serves as a trend indicator, rather than just a number.

c) Country Analysis - Depth vs Quality Pattern

Overview:

- The dataset covers 64 countries, showing wide global participation in professional tennis.
- Leading Nation: USA. With 50 competitors, the USA leads in participation numbers.
- Avg Points per Country: 636.27. The global average serves as a standard for comparing nations.

Top countries by competitor:

- USA clearly leads in total competitors, followed by France and Great Britain.
- There is a gradual drop after the top 3 - 4 countries, showing participation concentration.

Top countries by average points:

- El Salvador, Monaco and Finland show high average points despite very low competitor counts.
- Great Britain appears in both the participation top list and the high average points, indicating both depth and strength.

Competitors vs average points:

- No strong positive correlation between competitor count and average points is visible.
- Countries like Great Britain stand out as both deep and strong.
- Countries with many competitors (like USA) sit in mid-high average range, not at the top.

Country share of Average points:

- El Salvador have 34.7% of the average points share. It dominates globally despite its small size.
- Monaco & Finland combined close to 25%. This supports the star-driven narrative.
- Great Britain reaches 8.16%. It contributes through both depth and quality.

d) Leaderboards Page

- Great Britain dominates the podium with two players in the top three. This indicates strong national depth.
- Finland's presence shows that smaller nations can still produce elite players.
- The point difference between #1 and #2 is small (150 points), indicating fierce competition at the top.

Top 10 Players by Points:

- Clustered Strength: Great Britain has 5 players in the top 10, confirming its dominance.
- Star Nations: El Salvador (Arevalo, Marcelo) and Croatia (Pavic, Mate) have standout players, showing dependence on stars instead of depth.

Biggest Ranking Movers:

- The USA is developing multiple rising players. This suggests a strong talent pipeline.
- Smaller nations (Colombia, Denmark, Serbia, Turkey, Chinese Taipei) appear in the movers list, showing that new talent spreads globally.

Country Strength:

- Great Britain's strength is both top-heavy (podium players) and broad (multiple top 10 entries).
- The USA combines depth and growth, making it a key talent hub.
- European nations (France, Germany, Spain, Italy) form a robust competitive block.

e) Search & Filter Page

The filter page allows filtering by:

- Ranking range
- Points threshold
- Movement
- Country

Filters help find specific profiles, like rising players or consistent performers.

- Depth Nations vs Star Nations: Great Britain has a steady talent stream, while countries like El Salvador depend on a few star players.
- Performance vs Participation: Recruiters can tell the difference between active but average players and those who are both effective and elite.

f) SQL Query Page - Structural Data Insight

The page displays the SQL code and query results side by side.

7. Conclusion:

This project transforms raw SportRadar tennis API data into a structured MySQL database and an interactive Streamlit analytics dashboard. It enables meaningful analysis of player rankings, country performance, and competition trends through a well-designed schema and SQL views. Overall, it demonstrates a complete practical workflow from data extraction to insight-driven visualization.